

② Support Vector Regression

Support vector regression is a method to predict numbers by fitting a line (or curve) that ignores small errors and only cares about big mistakes.

Instead of fitting exactly through points like in linear regression, SVR builds a tube around a line.

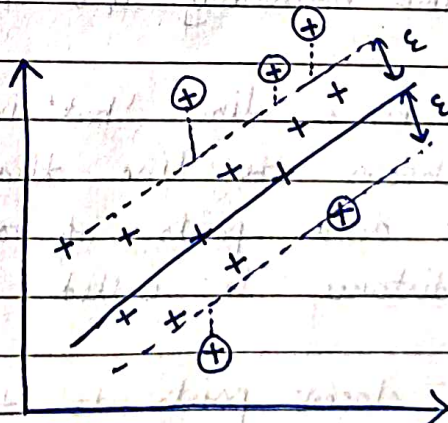
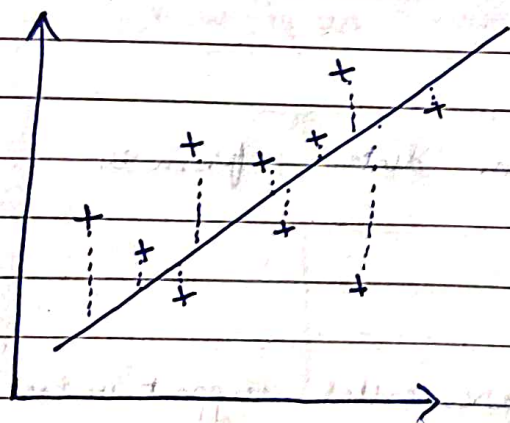
Points inside the tube $\Rightarrow$ no error

Points outside the tube $\Rightarrow$ penalty

Linear
Regression

SVR

- Fits all points
- Sensitive to noise
- No margin
- Ignores small errors
- Robust
- Has epsilon tube



In SVR the points that touch the tube boundary or the ones which lie outside the tube are called support vectors.